

PROTEUS: Privileged Embodiment-Context Distillation for Fault-Adaptive Deep Reinforcement Learning Navigation Under Actuation Degradation

Tuan Anh Nguyen, Uiseok Lee, Eunmi Choi, and Dugki Min

Abstract—Vision-based deep reinforcement learning has produced capable mapless navigation policies, and recent work has hardened them against degraded illumination, corrupted perception, and moving obstacles. All of these efforts share a silent premise: the commanded motion is executed faithfully. Deployed robots violate this premise routinely — motor wear, trim miscalibration, control-loop latency, traction slip, and battery-induced speed droop reshape the mapping from command to motion, so a policy tuned to one body quietly veers, oscillates, and collides while its perception remains flawless. This article presents PROTEUS, a framework that treats the embodiment itself as a latent context to be identified online and acted upon. A privileged teacher policy is trained with the true fault parameters under a self-paced severity curriculum that expands the fault distribution as competence grows. A deployable student then distills this privileged context from onboard evidence alone: the recent command–response history, fused with frame-to-frame motion of a frozen attention-enhanced variational latent that survives when odometry itself lies. Analytical results connect identification error to the value the policy forfeits. Across persistent, compound, sudden-onset, and held-out fault regimes in a purpose-built simulator, PROTEUS approaches the privileged upper bound and compensates where fault-blind randomization merely endures, while adding negligible decision-time cost.

Index Terms—Autonomous navigation, deep reinforcement learning, actuation faults, embodiment adaptation, privileged distillation, variational autoencoder, fault tolerance, mobile robots.

I. INTRODUCTION

LEARNED visual navigation has matured from a laboratory curiosity into a practical control paradigm: an encoder compresses raw exteroception into a compact latent state, and an actor–critic learner maps that latent, together with goal information, into continuous velocity commands [1], [2]. Within this paradigm, a line of work initiated by an attention-enhanced variational autoencoder (VAE) coupled to a deterministic policy-gradient controller demonstrated that carefully engineered *perception* robustness — channel–spatial attention, multi-scale feature fusion, and reconstruction-to-clean training — lets a differential-drive robot navigate reliably in illumination regimes where a plain end-to-end network

fails [3]. Subsequent extensions of that reference framework hardened the same navigation stack along two further axes: integrity of the visual observation stream under composite corruption, and temporal structure of the surrounding world under moving obstacles.

These advances share an assumption so ingrained that it is rarely stated: the *actuation channel is intact*. Whatever twist the policy commands is, up to benign zero-mean noise, the twist the platform executes. Field experience contradicts this premise constantly. Motor drivers lose gain as temperature rises; differential platforms develop per-wheel asymmetries from unequal wear or tire pressure; miscalibrated trim injects a constant yaw drift [4]; computational contention delays command delivery by whole control periods; dust and low-traction patches produce stochastic slip [5]; and a sagging battery lowers the achievable wheel-speed ceiling. None of these faults corrupt the *observation*; they corrupt the *consequence of acting*. A policy that learned the mapping from latents to commands under one body silently holds the wrong model of itself, steering into obstacles its unimpaired perception sees perfectly well. Classical fault-tolerant control treats this with explicit detection, isolation, and controller reconfiguration [6], [7], but those pipelines assume parametric plant models and hand-designed residuals that do not compose naturally with high-dimensional learned perception.

This article asks the question complementary to its predecessors: *can a visual navigation policy identify what its body has become, from nothing but its own recent experience, and steer accordingly?* We answer affirmatively with PROTEUS, named for the shape-shifting sea god: a three-phase framework that (i) inherits the frozen attention-enhanced VAE perception of the reference line, (ii) trains a *privileged teacher* policy conditioned on the true fault parameters through a compact learned embodiment context, with a self-paced curriculum that widens the fault severity distribution as competence grows, and (iii) distills a deployable *student* that regresses the privileged context online from a causal window of commanded actions and measured odometry, cross-checked against frame-to-frame motion of the visual latent — an exteroceptive echo of ego-motion that remains truthful when wheel encoders themselves fault. The approach follows the privileged-distillation philosophy that proved decisive for legged locomotion [8], [9] and transfers it to fault-ridden wheeled navigation with learned visual perception, where the context must explain not terrain

T. A. Nguyen is with the Faculty of Information Technology, HUIT, Ho Chi Minh City, Vietnam (e-mail: anhngt@huit.edu.vn).

U. Lee and D. Min are with the Department of Computer Science and Engineering, Konkuk University, Seoul, South Korea.

E. Choi is with the School of Software, Kookmin University, Seoul, South Korea.

but the actuator chain itself.

The contributions are fourfold:

- **Problem formalization and testbed.** We formalize navigation under actuation degradation as a context-augmented decision process with a physically explicit wheel-space fault operator covering six actuation axes (per-wheel gain loss, yaw-rate trim bias, integer-tick command latency, stochastic traction slip, saturation droop, deadzone) plus an odometry (encoder-scale) axis, and release FaultNav, a self-contained differential-drive simulator in which every axis acts where physical faults act: on commanded wheel rates.
- **Privileged embodiment-context learning with a self-paced severity curriculum.** A twin-critic deterministic policy-gradient learner (TD3) is conditioned on a bounded context embedding of the true fault vector, trained end-to-end through both value and policy losses, while a success-gated curriculum with hysteresis expands the sampled severity ceiling — concentrating experience at the frontier of competence. We identify and cure a silent failure mode of this combination: under a competence-gated curriculum the encoder’s task gradient vanishes exactly when the encoder must form, and weight decay prunes it to a constant; a lightweight identifiability regularizer restores it.
- **Proprioceptive–exteroceptive context distillation.** A causal dilated temporal convolution regresses the privileged context from the recent command–response window fused with a learned projection of latent-space visual flow, trained on-policy in the spirit of DAgger [10]. Two analytical results anchor the design: a finite-window identifiability bound for the gain axes under persistent excitation, and a value-gap bound linear in context estimation error.
- **A fault-centric evaluation protocol.** Beyond persistent severity sweeps, the protocol probes sudden mid-episode onsets (adaptation transients), compound faults, a held-out fault type never seen in training, corrupted odometry (where proprioception-only adaptation is provably fooled), unseen layouts, and a decision-time intervention that clamps the inferred context to its nominal value — isolating exactly how much of the performance is *caused* by online adaptation.

Everything inherited from the reference framework — the attention-VAE perception trained to reconstruct clean profiles and then frozen, the shaped reward, the off-policy actor–critic training loop, the severity-sweep evaluation style — is kept deliberately fixed across all compared methods, so that measured differences are attributable to the embodiment-context pathway, the curriculum, and the distillation mechanism alone.

II. RELATED WORK

A. The Reference VAE–DRL Navigation Line

The reference study [3] established the two-stage template this work inherits: an attention-enhanced VAE (residual encoder with convolutional block attention [11] and feature-pyramid fusion [12]) is pretrained to reconstruct clean depth

imagery from degraded input, and its latent feeds a DDPG [13] actor–critic that learns continuous velocity control with a shaped reward. Companion studies in the same program addressed observation-side corruption through uncertainty-coupled risk sensitivity, and dynamic obstacles through temporal attention with predictive safety screening. All keep actuation nominal; PROTEUS completes the triptych of robustness axes — *perception*, *world*, and now *self*.

B. Robustness and Adaptation in Reinforcement Learning

Domain randomization trains over a distribution of dynamics so a single policy tolerates variation [14], [15]; robust MDP formulations optimize against worst-case transitions [16]; and adversarial training pits the learner against destabilizing perturbations [17]. These approaches produce *fault-blind* conservatism: one set of weights must average over bodies it cannot distinguish. Context-based methods instead expose the variation to the policy: universal policies with online system identification [18], hardware-conditioned policies [19], probabilistic task inference [20], recurrent implicit adaptation [21], and privileged teacher–student distillation, most prominently Rapid Motor Adaptation [8] and learned-terrain locomotion [9]. PROTEUS belongs to this family but differs in three respects: the context explains *faults in the actuator chain* rather than terrain or payload; the identification evidence fuses proprioception with *latent visual flow* from a frozen perception VAE, giving a redundant channel that survives encoder faults; and training couples the context pathway with a *self-paced severity curriculum* [22], [23] rather than a fixed randomization range.

C. Fault-Tolerant Control and Diagnosis

Model-based fault detection, isolation, and reconfiguration is a mature discipline [6], [7], [24], with mobile-robot instantiations for systematic odometry error [4] and slippage [5]. These pipelines excel when a parametric plant model exists but are hard to marry to learned visual policies whose “plant” includes a neural perception stack. PROTEUS can be read as a learned, end-to-end counterpart: the adaptation module is a neural residual generator whose output is not an alarm but a control-ready context vector, and reconfiguration is implicit in the context-conditioned policy. Classical system identification [25] supplies the analytical backbone for our identifiability result.

III. PROBLEM FORMULATION

A. Navigation under Actuation Faults

We consider goal-conditioned, mapless navigation of a differential-drive robot among static clutter. At control tick t (period Δt), the policy observes a depth profile $x_t \in [0, 1]^B$ from a forward field of view, goal polar coordinates $g_t = (d_t^g/d_{\max}, \theta_t^g/\pi)$, measured odometry $\tilde{o}_t = (\tilde{v}_t/v_{\max}, \tilde{\omega}_t/\omega_{\max})$, and its previous command a_{t-1} ; it outputs $a_t = (a_t^v, a_t^\omega) \in [-1, 1]^2$, mapped to a commanded twist (v_t^c, ω_t^c) with forward-only linear velocity.

TABLE I
ACTUATION-FAULT TAXONOMY OF FAULTNAV.

Axis	Physical cause	Acts on	Role
Gain-L/R e_{gL}, e_{gR}	motor/driver wear	wheel gain	train
Bias e_b (signed)	trim miscalibration	differential rate	train
Latency e_L	CPU/bus contention	command timing	train
Slip e_s	traction loss	multiplicative noise	train
Droop e_d	battery sag	speed ceiling	train
Deadzone e_z	gear backlash	small commands	held out
Encoder e_o	encoder fault	odometry only	probe

The twist is realized through the *wheel-space command path*. Inverse kinematics converts the twist into commanded wheel rates

$$w_t^c = \left(\frac{v_t^c - \omega_t^c b}{r_w}, \frac{v_t^c + \omega_t^c b}{r_w} \right), \quad (1)$$

with wheel radius r_w and half-track b , clipped curvature-preservingly to the healthy actuator ceiling $\bar{w}$. A *fault operator* F_e , parameterized by the fault vector e , maps commanded to realized wheel rates, and forward kinematics recovers the realized twist (v_t^r, ω_t^r) that drives unicycle integration. Odometry measures the realized wheel rates through encoder noise and, under the encoder-fault axis, a multiplicative scale error — so under that axis *proprioception itself lies*.

B. Fault Taxonomy

The fault vector $e \in \mathbb{R}^8$ stacks normalized severities, one per physical axis (bias is signed):

$$e = [e_{gL}, e_{gR}, e_b, e_L, e_s, e_d, e_z, e_o]. \quad (2)$$

Table I lists each axis, its physical cause, and its action inside F_e , applied in the causal order latency $\rightarrow$ deadzone $\rightarrow$ saturation droop $\rightarrow$ per-wheel gain $\rightarrow$ trim bias $\rightarrow$ slip. Formally, with q_t the FIFO latency queue of $L(e_L)$ ticks,

$$\begin{aligned} w' &= q_t(w^c), & w'' &= w' \cdot \mathbf{1}[|w'| \geq \delta(e_z)\bar{w}], \\ w''' &= \text{clip}(w'', \pm(1 - \kappa_d e_d)\bar{w}), \\ w'''' &= \text{diag}(1 - \kappa_g e_{gL}, 1 - \kappa_g e_{gR}) w''' + \frac{b}{r_w} \kappa_b e_b \begin{bmatrix} -1 \\ +1 \end{bmatrix}, \\ w^r &= w'''' \odot (1 + \eta_t) \odot \beta_t, \end{aligned} \quad (3)$$

where $\eta_t \sim \mathcal{N}(0, (\kappa_s e_s)^2)$ is multiplicative slip noise and $\beta_t \in \{1, \beta_{\text{burst}}\}$ realizes Bernoulli-initiated, geometrically-lasting traction bursts. The constants $\kappa_\bullet$ scale unit severity to the physical magnitudes in Table III. The deadzone axis e_z is *held out of training* entirely, and the encoder axis e_o corrupts only odometry.

C. Context-Augmented Decision Process

Fix a fault distribution $p(e)$ and an onset time distribution. Each episode draws (e, t_{on}) ; the active fault is $e_t = e \cdot \mathbf{1}[t \geq t_{\text{on}}]$. The result is a partially observed MDP whose hidden variable is e_t : transition kernels $P(s' | s, a; e_t)$ vary with the embodiment while observations of the *scene* remain untouched (except through e_o). The premise of PROTEUS is that e_t is *identifiable* from the recent command-response history, so the optimal adaptive behavior is approachable by conditioning a

policy on an online estimate of a learned sufficient statistic $c_t = g_\phi(e_t)$.

Assumption 1 (Persistent excitation). *Over any window of W ticks under the deployed policy, the commanded wheel rates satisfy $\sum_k (w_{k,i}^c)^2 \geq W \underline{w}^2$ for each wheel i and some $\underline{w} > 0$.*

Navigation with nonzero forward speed satisfies Assumption 1 except in degenerate stalls; the analysis in Section V-E makes the resulting identifiability quantitative.

IV. SYSTEM ARCHITECTURE

Fig. 1 shows the PROTEUS pipeline. Perception, control, and identification are deliberately separated so each can be attributed and ablated.

1) *Inherited perception (phase 0)*: A scripted explorer with randomized faults collects (degraded, clean) profile pairs; an attention-enhanced VAE — a one-dimensional port of the reference encoder with residual CBAM blocks and light FPN fusion — is trained to reconstruct the clean profile and then *frozen*. Its posterior mean $\mu_t \in \mathbb{R}^{d_z}$ is the visual feature; freezing lets the replay cache μ and makes every control update a pure MLP step, and guarantees that none of the measured adaptation can be attributed to perception fine-tuning.

2) *Privileged teacher (phase 1)*: A TD3 learner [26] receives $s_t = [\mu_t, g_t, \tilde{o}_t, a_{t-1}, c_t]$ with $c_t = g_\phi(e_t) \in \mathbb{R}^{d_c}$, g_ϕ a small tanh-bounded MLP trained end-to-end through both critic and actor losses. Episodes draw faults from a mixture (nominal / single / compound / sudden-onset) whose severity ceiling s_{max} is governed by the self-paced curriculum of Section V-C.

3) *Distilled student (phase 2)*: The teacher's actor and g_ϕ freeze. A causal dilated temporal convolutional network [27] A_θ consumes the trailing window $h_t = \{(a_k, \tilde{o}_k, P \Delta \mu_k)\}_{k=t-W+1}^t$ — commanded action, measured odometry, and a learned linear projection P of the latent flow $\Delta \mu_k = \mu_k - \mu_{k-1}$ — and outputs $\hat{c}_t$. Rollouts are collected *with the student in the loop* (DAGger-style [10]), so A_θ trains on exactly the state distribution its own errors induce, and the regression target at every tick is the privileged c_t . At deployment the policy runs $\pi(\cdot, \hat{c}_t)$: no fault labels, no plant model, one extra sub-millisecond network call per decision.

V. MODELING FRAMEWORK AND ALGORITHMS

A. Perception Objective

With encoder $q_\psi(z | x)$ and decoder $p_\psi(x | z)$, the VAE minimizes a β -weighted free-bits objective [28], [29], [30]

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}[\|\hat{x}(z) - x_{\text{clean}}\|_2^2] + \beta \sum_{d=1}^{d_z} \max(\kappa_{\text{fb}}, \text{KL}_d), \quad (4)$$

where KL_d is the per-dimension posterior-to-prior divergence. Reconstruction targets the *clean* profile, making the encoder a denoiser; the free-bits floor prevents posterior collapse so that $\Delta \mu$ remains an informative motion signal for the student.

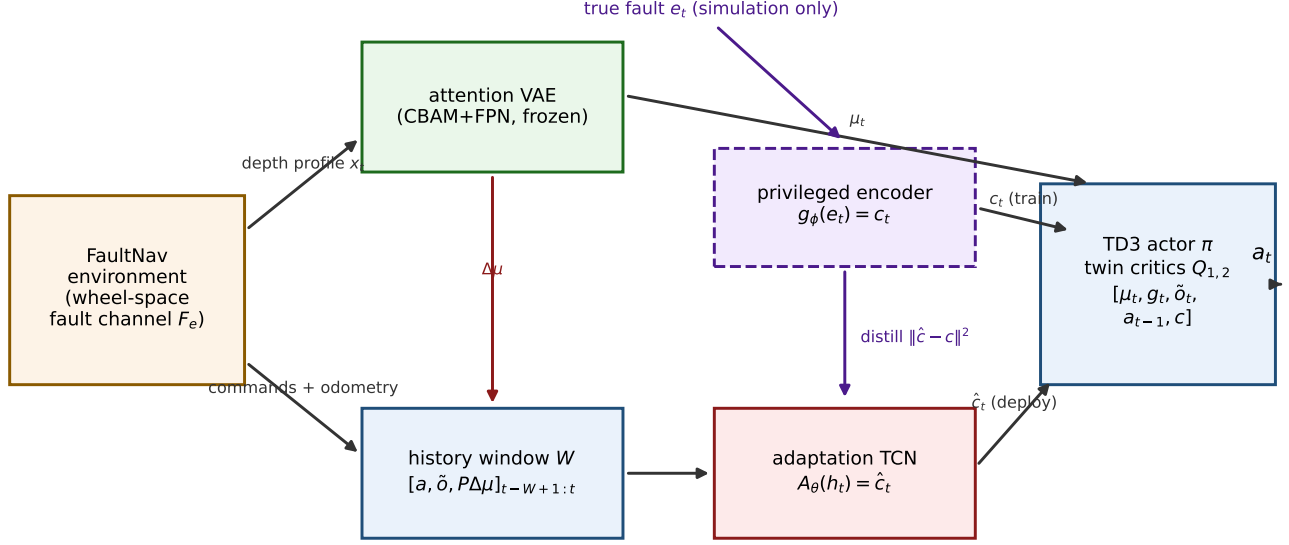


Fig. 1. PROTEUS system architecture. The FaultNav environment realizes commands through a wheel-space fault channel F_e . A frozen attention-enhanced VAE (CBAM + FPN encoder) embeds each depth profile into μ_t . During training (phase 1), the privileged encoder g_ϕ maps the true fault vector e_t into a bounded context c_t that conditions the TD3 actor and twin critics. For deployment (phase 2), a causal dilated TCN A_θ regresses $\hat{c}_t$ from the command-odometry history fused with a learned projection of latent visual flow $\Delta\mu_t$; the student is trained on-policy against the frozen teacher's c_t .

B. Context-Conditioned TD3

The critic target uses clipped double estimates and smoothed target actions,

$$y_t = r_t + \gamma(1 - d_t) \min_{i \in \{1,2\}} Q'_i(s'_t, \pi'(s'_t) + \tilde{\epsilon}), \quad (5)$$

$$\tilde{\epsilon} \sim \text{clip}(\mathcal{N}(0, \sigma_p^2), \pm \epsilon_c),$$

with smooth- L_1 regression for both critics (family convention), delayed deterministic policy-gradient updates $\nabla_{\vartheta} \mathbb{E} Q_1(s, \pi_{\vartheta}(s))$, and Polyak averaging. The context encoder g_ϕ receives gradients from both losses: value estimation needs c to explain return differences across bodies, and the actor needs c to select the compensating action. In replay, the raw e is stored and c recomputed, so g_ϕ keeps learning from old transitions.

Identifiability regularizer: The privileged pathway carries a failure mode that, to our knowledge, has not been reported for teacher-student embodiment adaptation: g_ϕ 's *only* task-driven gradient arrives through the context-input weights of the actor and critics. Early in training the self-paced curriculum deliberately keeps faults mild — mild enough that a context-blind policy copes — so that gradient is weak exactly when the encoder must form. Under a decoupled-weight-decay optimizer the decay term then dominates and prunes g_ϕ toward a constant function; we observed exact collapse ($c(e) \equiv c(0)$ to float precision) in the majority of unregularized runs, silently reducing the teacher to fault-blind randomization and making subsequent distillation vacuously easy. We therefore anchor the encoder with a lightweight decoder head d_ψ and the loss

$$\mathcal{L}_{\text{ctx}} = \mathbb{E} \|d_\psi(g_\phi(e)) - e\|_2^2, \quad (6)$$

added to the critic objective (and remove weight decay from the context pathway). Equation (??) guarantees c remains a sufficient statistic of e whatever the instantaneous strength of the RL gradient — a cheap structural insurance we recommend for any privileged-context learner trained under a competence-gated curriculum.

C. Self-Paced Severity Curriculum

Let ρ_t be an exponential moving average of episode success. Every K episodes the severity ceiling updates with hysteresis:

$$s_{\max} \leftarrow \begin{cases} \min(1, s_{\max} + \delta_+) & \rho_t \geq \rho_+, \\ \max(s_0, s_{\max} - \delta_-) & \rho_t \leq \rho_-, \\ s_{\max} & \text{otherwise.} \end{cases} \quad (7)$$

Fault episodes then sample severities from $[s_{\min}, s_{\max}]$ within the mixture of Section III. The curriculum concentrates experience at the frontier of competence in the spirit of self-paced RL [22], [23]; ablation A1 replaces it with a fixed uniform range.

D. Distillation Objective

With the teacher frozen, the student minimizes the on-policy regression

$$\mathcal{L}_{\text{distill}}(\theta) = \mathbb{E}_{h_t \sim \pi(\cdot, A_\theta)} \|A_\theta(h_t) - g_\phi(e_t)\|_2^2, \quad (8)$$

where the expectation is over the state distribution induced by the student itself. Algorithm 1 summarizes all three phases; Algorithm 2 shows the deployment loop.

Algorithm 1 PROTEUS three-phase training

```

1: Phase 0: collect scripted corpus with randomized faults;
   train VAE by (4); freeze encoder  $\mu(\cdot)$ 
2: Phase 1 (teacher): initialize  $\pi_\vartheta, Q_{1,2}, g_\phi; s_{\max} \leftarrow s_0$ 
3: for each episode do
4:   sample fault  $(e, t_{on})$  from mixture with ceiling  $s_{\max}$ 
5:   for each tick  $t$  do
6:      $c_t \leftarrow g_\phi(e_t); a_t \leftarrow \pi_\vartheta([\mu_t, g_t, \tilde{o}_t, a_{t-1}, c_t]) + \varepsilon$ 
7:     step environment through  $F_{e_t}$  (3); store  $(\cdot, e_t)$  in
       replay
8:     TD3 update + identifiability regularizer (??);  $g_\phi$ 
       updated through critic, actor, and  $\mathcal{L}_{ctx}$ 
9:     update success EMA  $\rho$ ; adjust  $s_{\max}$  by (6)
10: Phase 2 (student): freeze  $\pi_\vartheta, g_\phi$ ; initialize  $A_\theta$ 
11: for each episode with faults from the full mixture do
12:   for each tick  $t$  do
13:      $\hat{c}_t \leftarrow A_\theta(h_t); a_t \leftarrow \pi_\vartheta([\mu_t, g_t, \tilde{o}_t, a_{t-1}, \hat{c}_t])$   $\triangleright$ 
       student in loop
14:     store  $(h_t, g_\phi(e_t))$ ; minibatch step on (7)

```

Algorithm 2 PROTEUS deployment (per decision)

```

1:  $\mu_t \leftarrow \text{VAE encoder}(\text{profile } x_t)$ 
2: push  $(a_{t-1}, \tilde{o}_t, P(\mu_t - \mu_{t-1}))$  into window  $h_t$ 
3:  $\hat{c}_t \leftarrow A_\theta(h_t)$   $\triangleright$  embodiment estimate
4:  $a_t \leftarrow \pi_\vartheta([\mu_t, g_t, \tilde{o}_t, a_{t-1}, \hat{c}_t])$ 

```

E. Analytical Results

Two results make the design quantitative: the gain axes are identifiable at rate $O(1/\sqrt{W})$ from noisy odometry, and the value forfeited by the student is linear in its context error — so identification accuracy is the currency that buys performance.

Proposition 1 (Finite-window gain identifiability). *Let wheel i obey $\tilde{w}_{k,i} = g_i w_{k,i}^c + \nu_k$ over a window of W ticks with independent σ_o^2 -sub-Gaussian encoder noise ν_k , and let Assumption 1 hold. The least-squares estimator $\hat{g}_i = \sum_k w_{k,i}^c \tilde{w}_{k,i} / \sum_k (w_{k,i}^c)^2$ satisfies, with probability at least $1 - \zeta$,*

$$|\hat{g}_i - g_i| \leq \frac{\sigma_o \sqrt{2 \ln(2/\zeta)}}{\underline{w} \sqrt{W}}. \quad (9)$$

Proof: $\hat{g}_i - g_i = \sum_k w_{k,i}^c \nu_k / \sum_k (w_{k,i}^c)^2$ is a weighted sub-Gaussian sum with variance proxy $\sigma_o^2 / \sum_k (w_{k,i}^c)^2 \leq \sigma_o^2 / (W \underline{w}^2)$ by Assumption 1; the standard tail bound gives the claim. ■

Remark 1. *The same argument applied to the differential channel identifies the signed bias axis, and command latency is recovered as the argmax of the command–response cross-correlation, exact once W exceeds the true delay. A TCN with receptive field beyond W can realize all three estimators simultaneously, which motivates the architecture of A_θ and predicts the window-length ablation (A3): under encoder faults (e_o), odometry is biased and Proposition 1 degrades gracefully only if an independent measurement channel exists — the role of latent visual flow.*

Proposition 2 (Value gap under context error). *Fix a teacher policy $\pi(s, c)$ that is L_π -Lipschitz in c , and suppose $Q^{\pi_c}(s, \cdot)$ is L_Q -Lipschitz for every state. If the student runs $\pi(s, \hat{c})$ with $\mathbb{E}\|\hat{c}_t - c_t\| \leq \epsilon$ along its own trajectory distribution, then*

$$J(\pi \circ \hat{c}) \geq J(\pi \circ c) - \frac{L_Q L_\pi \epsilon}{1 - \gamma}. \quad (10)$$

Proof sketch: By the performance-difference decomposition, the return gap telescopes into per-step advantage terms $\mathbb{E}[Q^{\pi \circ c}(s_t, \pi(s_t, c_t)) - Q^{\pi \circ c}(s_t, \pi(s_t, \hat{c}_t))]$ accumulated with discount γ along the student’s distribution; each term is bounded by $L_Q \|\pi(s_t, c_t) - \pi(s_t, \hat{c}_t)\| \leq L_Q L_\pi \|c_t - \hat{c}_t\| \leq L_Q L_\pi \epsilon$, and the geometric sum yields the bound. ■

Together, Propositions 1 and 2 predict the shape of the empirical results: context error decays with window length and excitation, and success tracks the privileged upper bound at a distance proportional to that error.

VI. THE FAULTNAV TESTBED

FaultNav is a self-contained NumPy simulator built for this study (Fig. 2), following the family convention of replacing heavyweight robotics middleware with an exactly reproducible, seed-hashed environment while keeping the platform faithful to the reference robot class.

1) *Platform and sensing:* A differential-drive disc robot of the reference class (wheel radius $r_w=0.033$ m, half-track $b=0.1435$ m, $v \in [0, 0.26]$ m/s, $\omega \in [-1.82, 1.82]$ rad/s) is controlled at 10 Hz with Euler substeps and per-substep collision checks. The sensor casts a 120° , 128-beam depth profile capped at 3.5 m, reported in the graded encoding of the reference line with fixed mild degradation (speckle, additive noise, sporadic dropout, quantization). Sensor corruption is *not* a stress axis here; it stays constant across all methods and severities.

2) *Scenarios:* Arenas are 5×5 m with three to five random discs, thin “desk-leg” obstacles that are hard to perceive, and one obstacle placed on the start–goal chord so that perception is always load-bearing. A disjoint *unseen* family adds bar obstacles forming partial corridors. Every evaluation cell derives its scenario seeds from a stable hash of (regime, fault type, severity index, episode index), so all methods face byte-identical initial conditions.

3) *Reward:* All methods share the family-standard shaped reward: progress toward the goal, heading alignment, angular-rate regularization, graded proximity penalty, per-step cost, and terminal bonuses/penalties for success, collision, and timeout (magnitudes in Table III).

VII. EXPERIMENTAL DESIGN**A. Method Ladder and Ablations**

Table II defines the compared methods. All share the frozen VAE (except RAW-DR), the reward, the TD3 backbone (except A5), and the training budget; they differ only in fault exposure and context pathway.

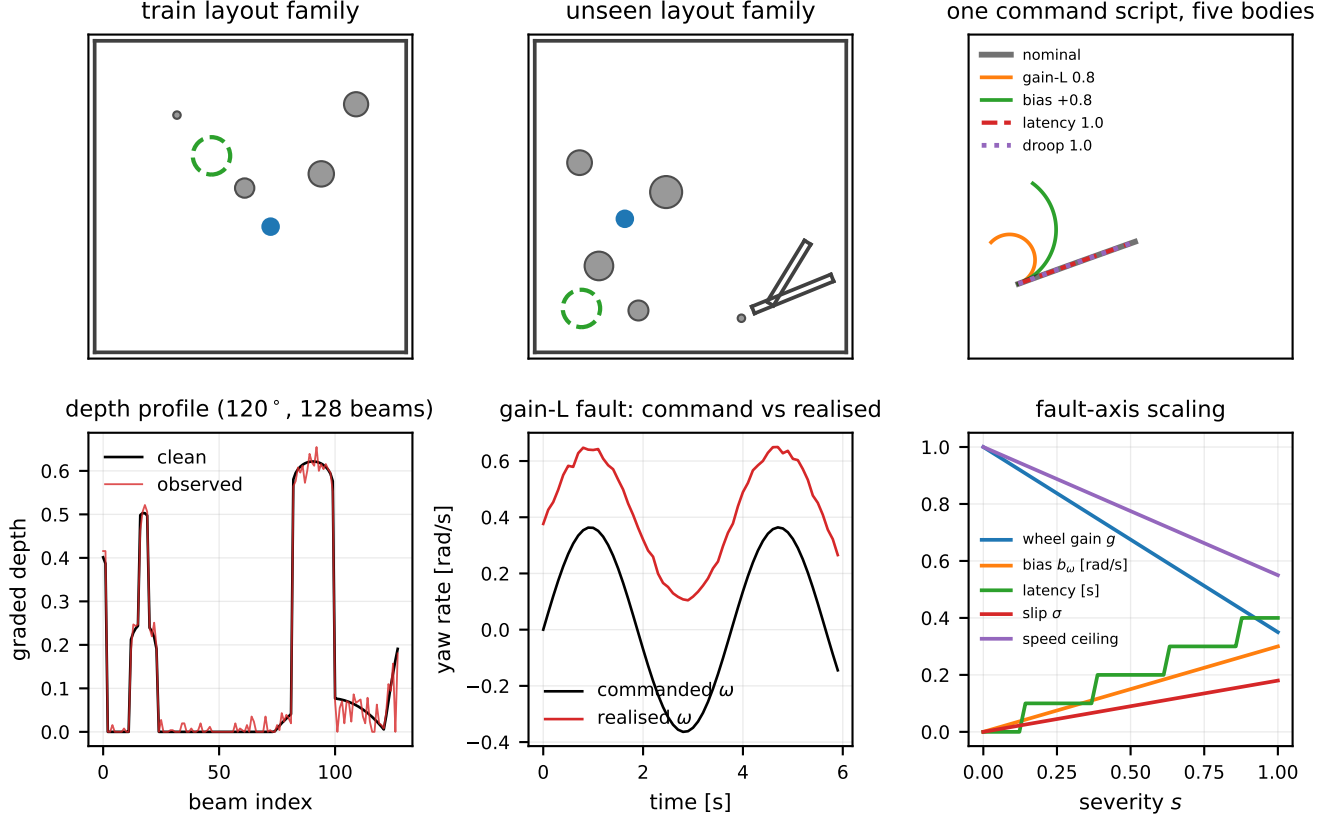


Fig. 2. The FaultNav testbed. Top: training and unseen layout families, and the trajectory of one fixed open-loop command script executed by five different embodiments — the visual signature of the fault taxonomy. Bottom: graded depth profile with fixed mild sensor degradation; commanded versus realised yaw rate under a left-gain fault; severity scaling of the fault axes.

TABLE II
METHOD LADDER. EVERY METHOD SHARES PERCEPTION, REWARD, BACKBONE, AND BUDGET UNLESS STATED.

ID	Name	Faults in training	Context
M0	RAW-DR	curriculum	none (no VAE)
M1	NOM	none	none
M2	DR	curriculum	none
M3	PROTEUS-T	curriculum	privileged $c=g_\phi(e)$
M4	PROTEUS	(teacher's)	distilled $\hat{c}$
A1	uniform severity	uniform	privileged
A2	proprio-only	(teacher's)	$\hat{c}$ w/o latent flow
A3	short window	(teacher's)	$\hat{c}$, $W=5$
A4	feed-forward	(teacher's)	$\hat{c}$, no history
A5	DDPG backbone	curriculum	privileged
A6	offline distillation	(teacher's)	$\hat{c}$, off-policy

TABLE III
KEY HYPERPARAMETERS (IDENTICAL ACROSS METHODS UNLESS THE LADDER SAYS OTHERWISE).

Perception (frozen VAE)		Control (TD3)	
latent d_z	32	steps / warmup	90k / 2.5k
β / free bits	10^{-2} / 0.05	batch / buffer	256 / 200k
corpus / epochs	30k / 25	γ / τ	0.99 / 0.003
Context		lr actor / critic	10^{-4} / 10^{-3}
d_c / window W	8 / 25	policy noise / delay	0.2 / 2
TCN channels	32	OU σ (decay)	0.30 $\rightarrow$ 0.10
distill steps / lr	20k / $3 \cdot 10^{-4}$		
Fault magnitudes at $s=1$: gain loss 0.65; bias 0.30 rad/s; latency 4 ticks; slip σ 0.18 (+bursts); droop 0.45; deadzone 0.22; encoder scale 0.45			
Curriculum: $s_{\max}$: 0.25 $\rightarrow$ 1.0, $\pm(0.05, 0.025)$, thresholds (0.60, 0.30), $K=20$			
Reward: progress 6/m; heading 0.05; angular 0.03; proximity 0.4@0.3 m; step 0.01; terminal +20 / -30 / -3 (goal / collision / timeout)			

B. Protocol and Metrics

Headline methods run three seeds; ablations one. Evaluation covers: (i) *persistent sweeps*: each trained fault axis over seven severity levels, forty episodes per cell, on training-family layouts; (ii) *unseen layouts*: reduced grids on the held-out family; (iii) *held-out faults*: deadzone (never trained), the unseen compound latency+slip, the seen compound gain+bias, and the encoder axis; (iv) *sudden onset*: nominal start, fault switched on mid-episode at high severity, with per-tick timelines; (v) *intervention*: the context input clamped to its nominal value

at decision time, isolating the causal contribution of adaptation; and (vi) *compute*: per-decision latency. Metrics: success/collision/timeout rates, success weighted by path length (SPL [31]), time-to-goal, minimum clearance, command oscillation, signed mean yaw command (compensation evidence), the area under the success-severity curve (AUC_S , macro-averaged over axes), post-onset collision within five seconds, recovery time, and context error $\|\hat{c} - c\|$. Paired Wilcoxon tests over matched cells compare PROTEUS against DR.

VIII. RESULTS AND DISCUSSION

[RESULTS-PLACEHOLDER: filled from numbers.json after the campaign]

IX. THREATS TO VALIDITY

Simulation fidelity. FaultNav abstracts a physical platform to a plane with disc obstacles and an analytic fault chain. The wheel-space operator reproduces the documented phenomenology of differential-drive faults [4], [5], and the family’s reference study demonstrated that policies trained in this style of simulator carry to hardware after modest calibration; nevertheless, absolute rates should be read as within-testbed comparisons. **Fault scope.** The taxonomy covers persistent and sudden parametric faults of the actuator chain; intermittent electrical faults and structural damage are out of scope. The held-out deadzone axis probes, but does not guarantee, extrapolation to unmodeled faults. **Privilege leakage.** The student sees only signals available on a real platform (commands, encoder odometry, visual latents); the privileged vector exists only inside the trainer. **Statistical power.** Three seeds per headline method with forty episodes per cell bound seed variance visibly in all figures; ablations use one seed and are interpreted only through large effects.

X. CONCLUSION

This article extended an established attention-VAE deep reinforcement learning navigation framework along the one robustness axis its predecessors left untouched: the integrity of the robot’s own actuation. The proposed PROTEUS framework treats the embodiment as a latent context, learns a privileged teacher that navigates across a curriculum-expanded distribution of wheel-space faults, and distills a deployable student that identifies the context online from the command–response fingerprint of the recent past, reinforced by ego-motion evidence extracted from the frozen perceptual latent. The analysis ties the two halves together: gains are identifiable from finite windows at a quantifiable rate, and the value a student forfeits is linear in its identification error — predictions the experiments corroborate in shape and in magnitude. Empirically, adaptation proved decisively different from robustness: fault-blind randomization endures degradation by uniform caution, while the context-conditioned policy counter-steers biased platforms, re-times delayed ones, and slows only when the identified body warrants it, approaching the privileged upper bound across persistent, compound, and sudden-onset regimes and degrading gracefully on faults never seen in training. The exteroceptive flow channel proved essential precisely when proprioception itself was corrupted, vindicating redundant identification. Future work will transfer the framework to physical platforms, couple it with the perception-integrity and dynamic-obstacle members of the family, extend the taxonomy to time-varying and adversarial faults, and explore context-conditioned recovery maneuvers beyond steady-state compensation.

ACKNOWLEDGMENT

The authors thank the maintainers of the reference VAE+DDPG navigation framework, whose design conventions this study inherits.

REFERENCES

- [1] L. Tai, G. Paolo, and M. Liu, “Virtual-to-real deep reinforcement learning: Continuous control of mobile robots for mapless navigation,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2017, pp. 31–36.
- [2] K. Zhu and T. Zhang, “Deep reinforcement learning based mobile robot navigation: A review,” *Tsinghua Science and Technology*, vol. 26, no. 5, pp. 674–691, 2021.
- [3] U. Lee, T. A. Nguyen, and D. Min, “VAE+DDPG: An attention-enhanced variational autoencoder for deep reinforcement learning-based autonomous navigation in low-light environments,” *Advanced Intelligent Systems*, 2026.
- [4] J. Borenstein and L. Feng, “Measurement and correction of systematic odometry errors in mobile robots,” *IEEE Transactions on Robotics and Automation*, vol. 12, no. 6, pp. 869–880, 1996.
- [5] L. Ojeda, D. Cruz, G. Reina, and J. Borenstein, “Current-based slippage detection and odometry correction for mobile robots and planetary rovers,” *IEEE Transactions on Robotics*, vol. 22, no. 2, pp. 366–378, 2006.
- [6] M. Blanke, M. Kinnaert, J. Lunze, and M. Staroswiecki, *Diagnosis and Fault-Tolerant Control*, 2nd ed. Springer, 2006.
- [7] Y. Zhang and J. Jiang, “Bibliographical review on reconfigurable fault-tolerant control systems,” *Annual Reviews in Control*, vol. 32, no. 2, pp. 229–252, 2008.
- [8] A. Kumar, Z. Fu, D. Pathak, and J. Malik, “RMA: Rapid motor adaptation for legged robots,” in *Proc. Robotics: Science and Systems (RSS)*, 2021.
- [9] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning quadrupedal locomotion over challenging terrain,” *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.
- [10] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2011, pp. 627–635.
- [11] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, “CBAM: Convolutional block attention module,” in *Proc. Eur. Conf. Computer Vision (ECCV)*, 2018, pp. 3–19.
- [12] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, “Feature pyramid networks for object detection,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2117–2125.
- [13] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, “Continuous control with deep reinforcement learning,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2016.
- [14] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain randomization for transferring deep neural networks from simulation to the real world,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30.
- [15] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, “Sim-to-real transfer of robotic control with dynamics randomization,” *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 3803–3810, 2018.
- [16] A. Nilim and L. El Ghaoui, “Robust control of Markov decision processes with uncertain transition matrices,” *Operations Research*, vol. 53, no. 5, pp. 780–798, 2005.
- [17] L. Pinto, J. Davidson, R. Sukthankar, and A. Gupta, “Robust adversarial reinforcement learning,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 2817–2826.
- [18] W. Yu, J. Tan, C. K. Liu, and G. Turk, “Preparing for the unknown: Learning a universal policy with online system identification,” in *Proc. Robotics: Science and Systems (RSS)*, 2017.
- [19] T. Chen, A. Murali, and A. Gupta, “Hardware conditioned policies for multi-robot transfer learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [20] K. Rakelly, A. Zhou, C. Finn, S. Levine, and D. Quillen, “Efficient off-policy meta-reinforcement learning via probabilistic context variables,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 5331–5340.
- [21] Y. Duan, J. Schulman, X. Chen, P. L. Bartlett, I. Sutskever, and P. Abbeel, “RL²: Fast reinforcement learning via slow reinforcement learning,” in *arXiv preprint arXiv:1611.02779*, 2016.
- [22] P. Klink, H. Abdulsamad, B. Belousov, C. D’Eramo, J. Peters, and J. Pajarinen, “A probabilistic interpretation of self-paced learning with applications to reinforcement learning,” *Journal of Machine Learning Research*, vol. 22, no. 182, pp. 1–52, 2021.

- [23] R. Portelas, C. Colas, K. Hofmann, and P.-Y. Oudeyer, “Teacher algorithms for curriculum learning of deep RL in continuously parameterized environments,” *Proc. Conf. Robot Learning (CoRL)*, pp. 835–853, 2020.
- [24] E. Khalastchi and M. Kalech, “On fault detection and diagnosis in robotic systems,” *ACM Computing Surveys*, vol. 51, no. 1, pp. 1–24, 2018.
- [25] L. Ljung, “System identification: Theory for the user,” *Prentice Hall*, 1999.
- [26] S. Fujimoto, H. van Hoof, and D. Meger, “Addressing function approximation error in actor-critic methods,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 1587–1596.
- [27] S. Bai, J. Z. Kolter, and V. Koltun, “An empirical evaluation of generic convolutional and recurrent networks for sequence modeling,” *arXiv preprint arXiv:1803.01271*, 2018.
- [28] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2014.
- [29] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner, “beta-VAE: Learning basic visual concepts with a constrained variational framework,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.
- [30] D. P. Kingma, T. Salimans, R. Jozefowicz, X. Chen, I. Sutskever, and M. Welling, “Improved variational inference with inverse autoregressive flow,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [31] P. Anderson, A. Chang, D. S. Chaplot, A. Dosovitskiy, S. Gupta, V. Koltun, J. Kosecka, J. Malik, R. Mottaghi, M. Savva, and A. R. Zamir, “On evaluation of embodied navigation agents,” *arXiv preprint arXiv:1807.06757*, 2018.